4. PROJECT DEVELOPMENT PHASE

Automating Employee Data Import & Relationship Mapping Using Import Sets and Dot Walking

1. DEVELOPMENT OBJECTIVE

This document is the technical implementation guide for the ServiceNow Automation Project. Its purpose is to provide the ServiceNow Administrator with a precise, reproducible set of configuration instructions for constructing all components necessary to fulfil the approved functional requirements. All requirements referenced herein are formally defined in the "2. Requirement Analysis" document; the corresponding milestone timeline is established in the "3. Project Planning Phase" document. This guide addresses the build activities only and does not revisit planning, stakeholder, or scheduling considerations.

The development activities in this document are organised into five discrete tasks, each aligned to a planning milestone. Each task includes a specific objective, a step-by-step configuration guide, expected outcomes, and developer notes where supplementary technical context is required. The tasks must be executed in the sequence presented, as each task introduces a dependency for the one that follows.

Upon successful completion of all five tasks, the following capabilities will be fully operational in the development environment: a validated Import Set staging table for employee data ingestion; a fully configured Transform Map with complete direct and relationship field mappings; confirmed employee User records in the [sys_user] table with resolved manager reference linkages; and an Incident form displaying three dot-walked fields that auto-populate from the assigned User record.

2. DEVELOPMENT ENVIRONMENT

All configuration activities described in this document must be performed within a dedicated development environment. The following table defines the technical context and access requirements against which

this guide was authored. Any deviation from these environment parameters should be reviewed with the Project Manager before build activities commence.

Instance Type	ServiceNow Personal Developer Instance (PDI) or dedicated Development Instance
Application Scope	Global scope. No application-specific scope is required for this configuration.
Required Platform Roles	admin (for Import Sets, Transform Maps, and table configuration); itil (for Incident form access and layout modification)
ServiceNow Release	Any generally available release supporting native Import Sets, Transform Maps, and dot-walking. No additional plugins are required.
Source Data File	employee data.csv — prepared with headers: name, email, department, manager, phone number
Target Table	sys_user — the core User table; must be accessible to the admin role
Form Target	Incident table [incident] — default Incident form view

3. DEVELOPMENT TASKS & TECHNICAL IMPLEMENTATION GUIDE

TASK 1 | *Milestone M1 — 1 Day*

Data Source Preparation

Objective: Prepare and validate the external employee data file that will serve as the source for the Import Set. The file must conform to the field specification defined in the Data Dictionary (Requirement Analysis, Section 6).

1.1 Required File Structure

The source data file must be a CSV or Excel document with the following exact column headers in the first row. Column header names are case-insensitive during import, but must be consistent across all import cycles to ensure Transform Map field mappings remain valid.

1.2 Configuration Steps

1	**Create the data file**	Open a spreadsheet application (Google Sheets, Microsoft Excel, or equivalent). In row 1, enter the five column headers exactly as shown above: name, email, department, manager, phone number. Populate a minimum of four employee records in rows 2–5.
2	**Populate manager values correctly**	`manager` Leave the manager column blank for any user who has no manager (e.g., top-level managers). A blank value will result in an empty reference field rather than an error.
3	**Save the file**	`employee data.csv` Alternative acceptable formats: .xlsx or .xml. CSV is preferred as it produces the most predictable column mapping behaviour during ServiceNow import. Confirm the saved file opens correctly and all five columns are present with no extra blank columns or merged cells.

TASK 2 | *Milestone M2 — 1 Day*

Import Set Configuration

Objective: Upload the prepared source file into ServiceNow via the Import Sets module to create the staging table and load the raw employee records into it for subsequent transformation.

2.1 Navigation Path

Navigation — ServiceNow Application Navigator

```
All  >  System Import Sets  >  Load Data
```

2.2 Configuration Steps

1	**Open the Load Data form**	Using the Application Navigator (top-left filter field), type "Load Data" and select it from the Import Sets module group. The Load Data form will open, presenting options to load data from a file or a data source.
2	**Select table creation mode**	In the "Import set table" field, select the option to create a new table rather than loading into an existing one. This instructs ServiceNow to auto-generate a new staging table based on the structure of the uploaded file.

3	**Configure the new staging table name**	`import employee`
		`u_u_import_employee`
		This prefix is applied automatically to all user-created tables in the Global scope and is expected behaviour.

4	**Upload the source file**	`employee data.csv` Confirm the file name is shown in the upload field before proceeding.
5	**Submit the load request**	Click Submit to initiate the file upload and table creation process. ServiceNow will parse the file, create the staging table with columns derived from the CSV headers, and load all data rows into the table. The platform will redirect to the Import Set record upon successful completion.
6	**Verify the staging table**	`u_import_employee` Confirm the record count matches the number of data rows in the source file (minimum four records). Verify that all five expected columns are present: name, email, department, manager, and phone number.

		If any column is missing, return to Step 1 and re-examine the source file headers before re-uploading.

TASK 3 | *Milestone M3 — 2 Days*

Transform Map Development

Objective: Create and fully configure the Transform Map that defines how data moves from the [u_import_employee] staging table to the target [sys_user] User table, including both auto-mapped direct fields and the manually configured Manager reference relationship.

3.1 Navigation Path

Navigation — ServiceNow Application Navigator

```
All  >  System Import Sets  >  Create Transform Map
```

3.2 Part A — Transform Map Header Configuration

1	**Open the Transform Map creation form**	Navigate to All > System Import Sets > Create Transform Map and click New to open a blank Transform Map record.
2	**Set the Transform Map name**	`Employee Import Transform` This name will be used to identify the Transform Map throughout the platform and in all related documentation.
3	**Set the Source Table**	`u_import_employee` This links the Transform Map to the staging table created in Task 2.
4	**Set the Target Table**	`User [sys_user]` This instructs the transform engine to write processed records into the core User table.
5	**Save the Transform Map header**	Click Submit to save the Transform Map header record. The platform will redirect to the saved Transform Map record. The Field Map related list will now be visible at the bottom of the form — it will be empty at this stage.

TASK 4 | *Milestone M4 — 1 Day*

Data Transformation Execution

Objective: Execute the configured Transform Map against the staged Import Set records to migrate employee data into the [sys_user] table and verify that all records are correctly inserted with resolved field relationships.

4.1 Configuration Steps

1	**Open the Transform Map record**	`Employee Import Transform` Confirm the source and target tables are correctly set and that all five Field Map records are present in the related list before proceeding.
2	**Initiate the Transform**	Locate the Related Links section at the top of the Transform Map form. `Transform` A dialogue or processing screen will appear. ServiceNow will begin processing each record in the [u_import_employee] staging table against the defined field mappings.
3	**Monitor transformation progress**	The platform will display a transformation progress indicator and, upon completion, a Transform History record. `Completed` Note the Import Set run identifier (e.g., ISET0010008) displayed on the Transform History record for traceability purposes. Review the Inserted, Updated, Ignored, and Error counts in the Transform History. For a clean first run against new records: Inserted should equal the number of rows in the source file; Errors should be zero.
4	**Navigate to [sys_user] and verify inserted records**	Navigate to All > User Administration > Users to open the [sys_user] list view. Search for each employee from the source file by name or email to confirm the corresponding User records have been created. Confirm all five fields (Name, Email, Department, Phone, Manager) are populated on each inserted record.
5	**Verify the Manager reference relationship**	Open one of the imported employee User records that has a manager specified in the source file. `Manager` Click the reference link in the Manager field. The platform should navigate to the manager's User record — this confirms the

relationship is correctly established as a system reference, not a text copy.

If the Manager field is blank or unresolved, refer to the Developer Note in Section 6.1 regarding manager record sequencing.

TASK 5 | *Milestone M5 — 1 Day*

Dot-Walking Configuration on Incident Form

Objective: Configure the Incident form layout to include three dot-walked fields that automatically display the assigned user's Department, Email, and Manager by traversing the reference relationship from the Incident record to the [sys_user] table.

5.1 Understanding Dot-Walking in this Context

Dot-walk field paths on the Incident form

```
assigned_to.department    →  The Department of the assigned user
assigned_to.email         →  The Email address of the assigned user
assigned_to.manager       →  The Manager reference of the assigned user
```

5.2 Navigation Path

Navigation — open any Incident record to access form configuration

```
All  >  Service Desk  >  Incidents  >  Create New   (then access form
configuration from within the form)
```

5.3 Configuration Steps

1	**Open a new Incident record**	Navigate to All > Service Desk > Incidents > Create New to open a blank Incident form. The form must be open and rendered in the browser to access the form layout configuration option.
2	**Access the Form Layout configuration**	Right-click on any empty area of the Incident form header, or click the form context menu icon (typically visible at the top of the form body). `Configure  >  Form Layout`

3	**Locate the dot-walked fields in the Available column**	In the Available fields column, scroll or search for the following fields: — Assigned to → Department — Assigned to → Email — Assigned to → Manager

4	**Move the fields to the Selected column**	Select each of the three fields in turn and use the right-arrow control (or double-click) to move them from the Available column to the Selected column. Position them within a logical section of the form, such as below the existing 'Assigned to' field, to maintain form readability. All three fields must be present in the Selected column before saving.

4. FIELD MAPPING SPECIFICATION

The following table is the authoritative field mapping specification for the Transform Map configured in Task 3. It must be used as the reference document during all Field Map configuration activities and as a verification checklist following auto-mapping execution. Any deviation from the mappings defined here — including changes to coalesce settings — must be reviewed and approved before implementation.

Source Field (Staging Table)	Target Field ([sys_user])	Mapping Type	Coalesce	Configuration Notes
`name`	`name`	Direct	No	Auto-mapped. Accepts the full display name of the employee as provided in the source file.
`email`	`email`	Direct	Yes (Recommended)	Auto-mapped. Email is the recommended coalesce field to prevent duplicate record creation on re-import. Set Coalesce to true on this Field Map record.
`department`	`department`	Direct	No	Auto-mapped. Source value must correspond to a Department record that exists in the platform for the reference to resolve correctly.

Source Field (Staging Table)	Target Field ([sys_user])	Mapping Type	Coalesce	Configuration Notes
`phone_number`	`phone`	Direct	No	May require manual creation if auto-map does not resolve the name difference between 'phone number' (source) and 'phone' (target).
`manager`	`manager`	**Relationship**	No	Requires manual Field Map creation. The source text value is resolved to a [sys_user] record reference. Manager records must exist in [sys_user] at the time of execution.

4. CONFIGURATION PARAMETERS QUICK REFERENCE

The following tables provide a consolidated quick-reference summary of all configuration parameter values used across Tasks 2 through 5. These values are definitive and must be applied exactly as specified. Any deliberate deviation — such as a different staging table name — will require corresponding updates to all downstream Transform Map references and should be documented as a change.

5.1 Import Set & Transform Map Parameters

Component	Parameter	Value / Setting
Import Set Table	Display Name	`import employee`
Import Set Table	Physical Table Name	`u_import_employee`
Import Set Table	Source File	`employee data.csv`
Transform Map	Name	`Employee Import Transform`
Transform Map	Source Table	`u_import_employee`
Transform Map	Target Table	`User [sys_user]`
Transform Map	Coalesce Field	`email` (recommended — set on Field Map record)

5.2 Incident Form Dot-Walk Parameters

Field Label (Form Display)	Dot-Walk Path	Source Reference Field	Data Type
Assigned to → Department	`assigned_to.department`	`assigned_to` (ref: sys_user)	Reference (Department)
Assigned to → Email	`assigned_to.email`	`assigned_to` (ref: sys_user)	Email (String)
Assigned to → Manager	`assigned_to.manager`	`assigned_to` (ref: sys_user)	Reference (sys_user)

6. DEVELOPER NOTES & BEST PRACTICES

6.1 Manager Record Sequencing and Resolution

The Manager field in [sys_user] is a reference field whose value is resolved at transformation time. If the named manager does not yet exist as a User record in [sys_user] when the Transform Map processes the dependent employee row, the manager reference will not be set. The developer must address this through one of the following approaches, selected based on the composition of the source file:

- **Approach A — Pre-existing Manager Records:** If all managers referenced in the source file already exist as User records in [sys_user], no special handling is required. The transform will resolve the reference automatically during execution.
- **Approach B — File Ordering:** If the source file contains both manager and employee records, place all manager rows at the top of the file before any rows that reference them. The transform processes rows sequentially, so managers inserted earlier in the run will be available for reference by later rows.
- **Approach C — Second Transform Run:** If file ordering is not feasible (e.g., the file is generated by an external process), execute the Transform Map twice against the same Import Set. The first run inserts all records; the second run resolves any previously unresolved manager references, as all User records will now exist in [sys_user].

6.2 Coalescing for Idempotent Re-Imports

Coalescing is the mechanism by which the Transform Map identifies whether an incoming record should be inserted as a new User record or used to update an existing one. Without a coalesce field configured, every execution of the Transform Map against the same Import Set data will insert duplicate records.

To enable idempotent re-import behaviour (safe to re-run without creating duplicates), open the email Field Map record within the 'Employee Import Transform' Transform Map and set the Coalesce toggle to true. With this configuration, the transform engine will search for an existing [sys_user] record whose email matches the incoming value before deciding whether to insert or update. The email field is the recommended coalesce field for this dataset as it is expected to be unique per employee.

7. DEVELOPMENT SIGN-OFF CRITERIA

The following criteria must all be satisfied and independently verified before the development phase is formally closed and the solution is handed over for end-to-end testing (Milestone M6). Each criterion must be confirmed by the ServiceNow Administrator who performed the configuration and countersigned by the designated QA Tester or Project Manager. Incomplete sign-off items must be remediated before testing commences.

#	Criterion	Verified By	Date
SC-01	The [u_import_employee] staging table exists in the ServiceNow instance and contains a minimum of four employee records with all five source columns (name, email, department, manager, phone number) correctly populated.		
SC-02	The 'Employee Import Transform' Transform Map exists and is correctly configured with [u_import_employee] as the source table and [sys_user] as the target table.		
SC-03	The Transform Map Field Maps related list contains five records covering all source-to-target field mappings as specified in Section 4 of this document, including the manually created Manager relationship mapping.		
SC-04	The email Field Map record has Coalesce set to true, enabling safe re-import behaviour without duplicate record creation.		

#	Criterion	Verified By	Date
SC-05	The Transform Map has been executed successfully and the Transform History record displays a state of 'Completed' with zero errors.		
SC-06	All imported employee User records are present and accessible in [sys_user], with all five fields (Name, Email, Department, Phone, Manager) correctly populated.		
SC-07	The Manager reference field on each applicable imported employee record resolves correctly to the corresponding manager's User record in [sys_user] and is navigable as a reference link.		
SC-08	The Incident form layout includes the three dot-walked fields: Assigned to Department, Assigned to Email, and Assigned to Manager, and these are visible on the form.		
SC-09	When an imported employee user is selected in the 'Assigned to' field of an Incident record, all three dot-walked fields auto-populate with the correct values from the corresponding [sys_user] record without requiring a page refresh or manual action.		
SC-10	All five tasks in this document have been completed in the correct sequence and all expected outcomes stated per task have been verified and documented.		